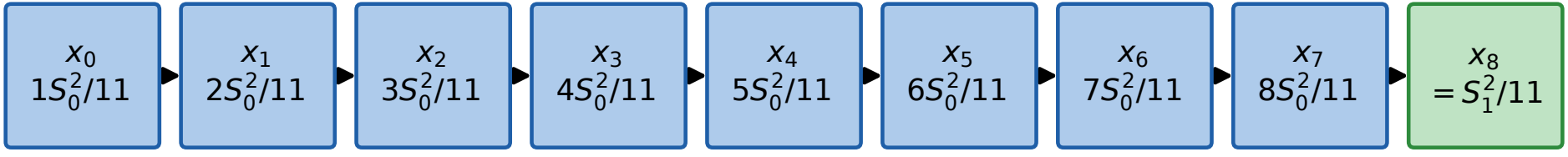
one period = 8 steps; $t = 1, \ldots, 9$; event at $t = 9$ ($u^* = 3$): $L : S_0 \to 3S_0$



$|D| = 7 \to 9$ at the event; D constant on the other steps